Master's Thesis Proposal

Title of the Study: Multiclass Skin Lesion Classification Using Deep Learning

A. Statement of the Problem

Skin cancer is one of the most common and life-threatening types of cancer, and early identification is important for increasing survival chances. Automated categorization of dermatoscopic pictures is a potential AI application, but existing methods have a significant limitation: dataset imbalance.

The HAM10000 dataset, widely used for skin lesion classification, contains 10,015 images across seven classes. The distribution is highly skewed: nevus (nv) has more than 6,700 samples, while classes like dermatofibroma (df) and vascular lesions (vasc) have fewer than 100. Standard deep learning models become biased toward the dominant class, leading to poor recall for minority yet clinically significant classes such as melanoma (mel).

- **Scope / Limitations:**Focused on HAM10000 dataset. Limited to dermatoscopic images only. No integration of other modalities (e.g., patient metadata beyond provided).
- **Hypothesis:** A combined oversampling and class-aware dynamic augmentation strategy will improve balanced accuracy and macro F1-score significantly compared to standard CNN training.
- **Major Assumptions:**
    1. HAM10000 labels are accurate.
    2. Features from ImageNet pre-trained CNNs are transferable to dermatoscopic images.
    3. Performance improvements on HAM10000 can serve as a proxy for clinical usefulness.

B. Justification for and Significance of the Study

The choice of this problem is motivated by the increasing burden of skin cancer worldwide and the urgent need for scalable diagnostic support systems. Current literature demonstrates CNN success in medical imaging but highlights limitations in generalizing to imbalanced datasets.

- **Relation to Literature:**
    - ResNet50 (He et al., 2016) showed improvements in image recognition but struggles on minority-heavy datasets.
    - DenseNet121 (Huang et al., 2017) introduced dense connectivity to improve efficiency.
    - EfficientNetB3 (Tan & Le, 2019) scales networks effectively and has shown state-of-the-art results on HAM10000.
    - HAM10000 dataset (Tschandl et al., 2018) is widely used but known for imbalance.
    - Esteva et al. (2017) showed CNNs can achieve dermatologist-level accuracy but acknowledged dataset bias.
- **Importance:**
    - To knowledge: This study tests data-centric imbalance strategies in a rigorous experimental setup.

- To healthcare: Provides groundwork for fairer AI models in dermatology.
- To the student: Builds advanced skills in deep learning, data augmentation, and experimental design.

C. Methodology or Procedures

1. Data Preparation

- **Dataset:** HAM10000 (10,015 images; 7 classes).
- **Splitting:** Stratified 70% training, 15% validation, 15% test.
- **Preprocessing:** Resize to 224×224, normalize pixel values, impute missing age values with the median.

2. Imbalance Mitigation

- **Oversampling:** Duplicate samples of minority classes (df, vasc, mel).
- **Dynamic Class-Aware Augmentation:**
  - Aggressive augmentation (rotation, cutout, zoom, contrast) for rare classes.
  - Lighter augmentation for dominant classes.

3. Model Training

- **Architectures:**
  - ResNet50 (baseline).
  - EfficientNetB3 (primary).
  - DenseNet121 (exploratory).
- **Procedure:**
  1. Feature Extraction (frozen backbone + custom head).
  2. Fine-Tuning (partial unfreeze, lower LR).
- **Optimization:** Adam/AdamW optimizers, callbacks (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint).

4. Evaluation Metrics

- **Primary:** Balanced Accuracy, Macro F1-score.
- **Secondary:** Overall accuracy, per-class recall, confusion matrices.
- **Interpretability:** Grad-CAM heatmaps for visual explanations.

D. Resources Required

- **Datasets:** HAM10000 (public).
- **Software:** TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, NumPy, pandas, matplotlib (all open-source).
- **Hardware:** Google Colab Pro (A100/T4 GPUs), local Jupyter environments.
- **Literature:** IEEE Xplore, ACM Digital Library, PubMed access via NYIT library portal.

E. Preliminary Results (from prior experiments)

- **ResNet50:** Accuracy 77.8%; Balanced Accuracy 0.78; Macro F1-score 0.78.
- **EfficientNetB3:** Accuracy 83.4%; Balanced Accuracy 0.817; Macro F1-score 0.825 (best performer).
- **DenseNet121:** Training in progress; expected to rival EfficientNetB3 after tuning.

These results support the hypothesis that **imbalance-aware augmentation** substantially improves minority class recall while maintaining stability.

F. Risks and Mitigation

- **Risk:** Overfitting to oversampled data.
  - Mitigation: Use on-the-fly augmentation, regularization (Dropout, Weight Decay), early stopping.
- **Risk:** Limited GPU resources.
  - Mitigation: Mixed precision training, efficient data pipelines.
- **Risk:** Dataset generalizability.
  - Mitigation: Perform stratified evaluation, recommend future work on external datasets.

G. Literature Cited

1. Esteva, A., et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639), 115–118.
2. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR*.
3. Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. *CVPR*.
4. Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for CNNs. *ICML*.
5. Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images. *Scientific Data*, 5(1), 1–9.

H. Revised Proposals

If significant changes arise — e.g., inclusion of additional datasets, adoption of alternative architectures, or a shift in evaluation strategy — a revised proposal will be submitted as required by NYIT.